1 Explanatory material

1.1 Calculating the linear correlation coefficient r

Cycle 2 – April 7th

What is the linear correlation coefficient?

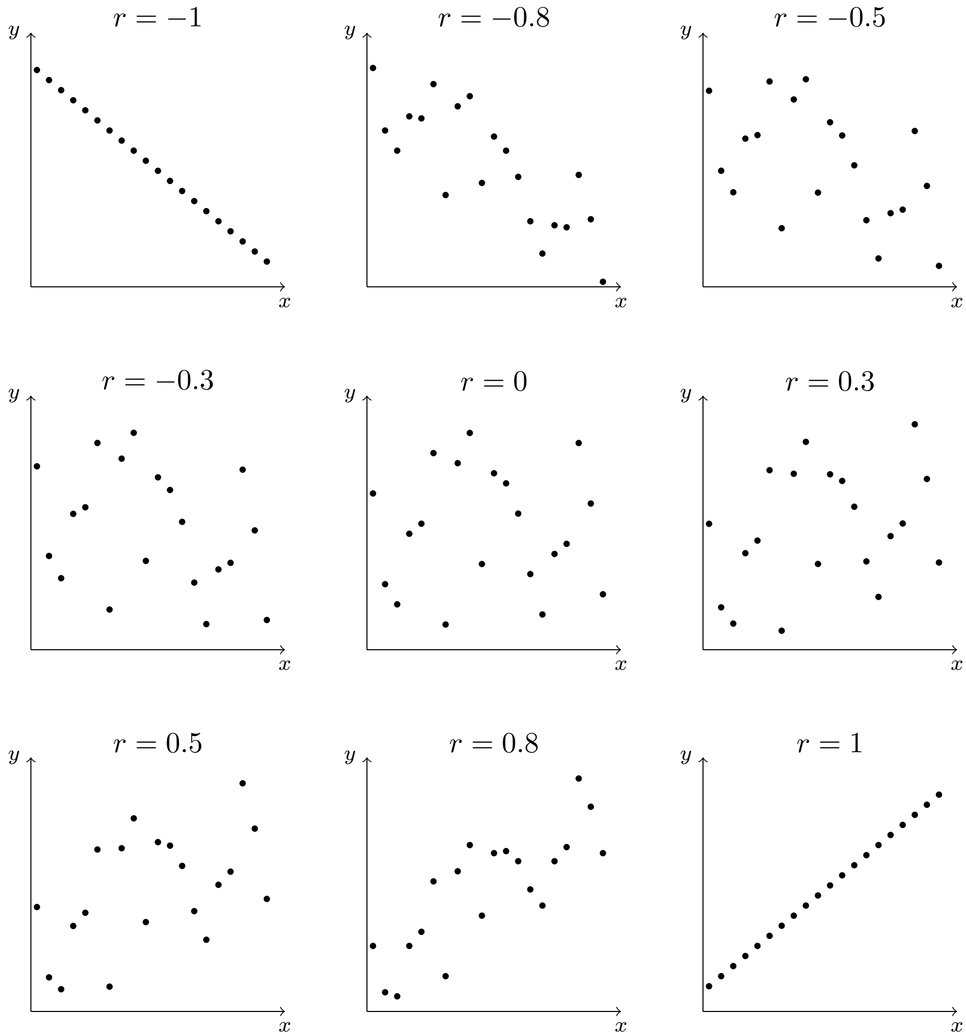
The linear correlation coefficient (r) measures the strength and direction of the relationship between two variables.

$$\begin{aligned} r = +1 &\rightarrow \text{perfect positive relationship} \\ r = 0 &\rightarrow \text{no linear relationship} \\ r = -1 &\rightarrow \text{perfect negative relationship} \end{aligned}$$

Examples:

- Positive correlation
- Negative correlation
- No correlation

Representative scatterplots



Linear Correlation Coefficient (r)

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2]\,[n \sum y^2 - (\sum y)^2]}}$$

Meaning of each part:

- r: The correlation coefficient. It ranges from -1 to $+1$.
- n: Number of data points (observations).
- $\sum x$: Sum of all x-values.
- $\sum y$: Sum of all y-values.

- $\sum xy$: Sum of the products of each x and y pair.
- $\sum x^2$: Sum of the squares of all x-values.
- $\sum y^2$: Sum of the squares of all y-values.

Interpretation of r:

- -1: Perfect negative correlation
- 0: No linear correlation
- $+1$: Perfect positive correlation

The closer r is to -1 or $+1$, the stronger the linear relationship.

Properties of the Linear Correlation Coefficient r

1. The value of r is always between -1 and 1 inclusive. That is,

$$-1 \leq r \leq 1$$

2. If all values of either variable are converted to a different scale, the value of r does not change.
3. The value of r is not affected by the choice of x or y. Interchange all x- and y-values and the value of r will not change.
4. r measures the strength of a linear relationship. It is not designed to measure the strength of a relationship that is not linear.
5. r is very sensitive to outliers in the sense that a single outlier can dramatically affect its value.

Why is it useful in real life?

Understanding correlation helps you:

- Make better decisions using data
- Understand how variables are connected in real situations
- Predict possible trends in economics and daily life
- Analyse information in studies, sport, and finance

1.2 Calculating the linear regression equation

Cycle 3 – April 9th

The linear regression equation using the least squares method

$$\hat{y} = a + bx$$

Where:

- $\hat{y}$ is the predicted value,
- a is the intercept,
- b is the slope,
- x is the independent variable.

Linear regression

To calculate the slope b:

$$b = \frac{n\sum(xy) - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

To calculate the intercept a:

$$a = \frac{\sum y - b\sum x}{n}$$

Relationship between the linear correlation coefficient and the regression slope

The linear correlation coefficient r and the regression slope b are related, but they are not the same quantity.

The linear correlation coefficient r measures the strength and direction of a linear relationship between two variables. The regression slope b in

$$\hat{y} = a + bx$$

measures how much the predicted value of y changes when x increases by 1 unit.

So, r is a measure of association, while b is a measure of change.

The formula for the linear correlation coefficient is

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}.$$

The formula for the regression slope is

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}.$$

Notice that the numerator is the same in both formulas:

$$n\sum xy - (\sum x)(\sum y).$$

However, the denominator is different. This is one reason why r and b are generally not equal.

The exact relationship between them is

$$b = r\frac{s_y}{s_x},$$

where s_y is the standard deviation of y and s_x is the standard deviation of x.

This formula shows that b depends not only on the direction and strength of the linear relationship, but also on the relative spread of the two variables. For this reason, r and b may sometimes look similar in value, but they are usually different.

It is also important to interpret them correctly:

- r is always between -1 and 1.
- r has no units.
- b is not restricted to the interval $[-1, 1]$.
- b has units, specifically units of y per unit of x.

In a specific example with nonzero correlation, if $r = b$, then

$$\frac{s_y}{s_x} = 1,$$

so the spread of x and y is the same. This is a special case, not a general rule.

A useful memory aid is:

- r answers: "How strong is the linear relationship?"
- b answers: "How much does the predicted y change when x increases by 1?"

For terminology, r should be called the linear correlation coefficient, and b should be called the regression slope or slope coefficient.

2 Activities and solutions

2.1 Task one

Complete the following table and calculate the Pearson product moment correlation coefficient. Draw the corresponding scatter plot.

(x)	(y)	x^2	y^2	xy
1	2	1	4	2
2	3	4	9	6
3	5	9	25	15
4	4	16	16	16
5	6	25	36	30
$\sum 15$	$\sum 20$	$\sum 55$	$\sum 90$	$\sum 69$

Solution

Using the values in the table, we have

$$n = 5, \qquad \sum x = 15, \qquad \sum y = 20, \qquad \sum x^2 = 55, \qquad \sum y^2 = 90, \qquad \sum xy = 69.$$

Substitute into the Pearson product moment correlation coefficient formula:

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}.$$

So,

$$r = \frac{5(69) - (15)(20)}{\sqrt{[5(55) - 15^2]\,[5(90) - 20^2]}}.$$

Now simplify:

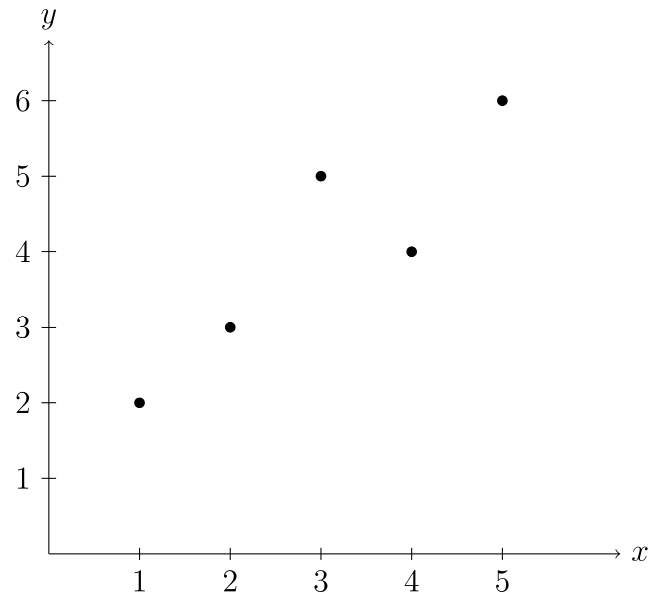
$$r = \frac{345 - 300}{\sqrt{(275 - 225)(450 - 400)}} = \frac{45}{\sqrt{50 \cdot 50}} = \frac{45}{50} = 0.9.$$

Therefore, the Pearson product moment correlation coefficient is

$$r = 0.9.$$

This indicates a strong positive linear correlation between x and y.

The corresponding scatter plot is shown below.



Now calculate the linear regression equation $\hat{y} = a + bx$.

First, compute the slope:

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2}.$$

Substitute the values:

$$b = \frac{5(69) - (15)(20)}{5(55) - 15^2} = \frac{345 - 300}{275 - 225} = \frac{45}{50} = 0.9.$$

Now compute the intercept:

$$a = \frac{\sum y - b\sum x}{n}.$$

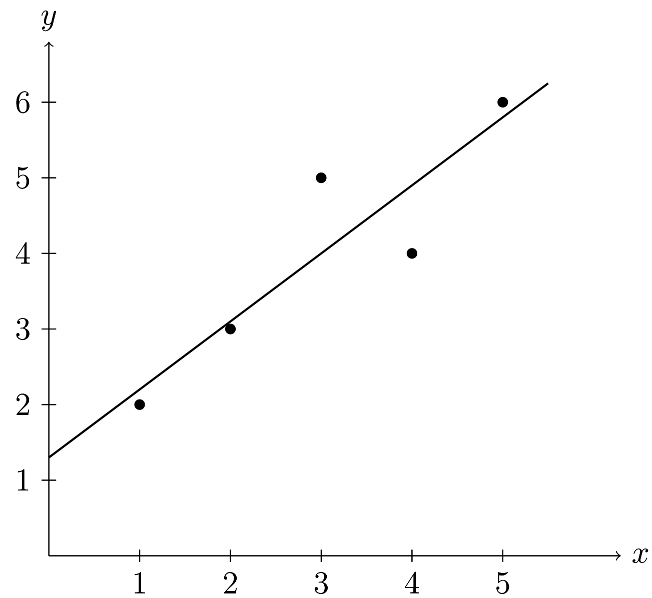
Substitute the values:

$$a = \frac{20 - (0.9)(15)}{5} = \frac{20 - 13.5}{5} = \frac{6.5}{5} = 1.3.$$

Therefore, the linear regression equation is

$$\hat{y} = 1.3 + 0.9x.$$

The regression line is shown below.



2.2 Warm up

Calculate the question mark value.

$$\begin{aligned} \text{accordion} + \text{accordion} + \text{accordion} &= 36 \\ \text{guitar} \times \text{guitar} + \text{accordion} &= 31 \\ \text{pair of guitars} + \text{pair of guitars} \times \text{pair of banjos} &= 190 \\ \text{accordion} + \text{banjo} \times \text{guitar} &= ? \end{aligned}$$

Solution

Interpreting each double-instrument icon as twice the corresponding single-instrument value:

$$3(\text{accordion}) = 36$$

so

$$\text{accordion} = 12.$$

From the second line,

$$(\text{guitar})^2 + 12 = 31$$

so

$$(\text{guitar})^2 = 19$$

and therefore

$$\text{guitar} = \sqrt{19}.$$

From the third line,

$$2(\text{guitar}) + 2(\text{guitar}) \cdot 2(\text{banjo}) = 190.$$

Substituting guitar $= \sqrt{19}$,

$$2\sqrt{19} + 4\sqrt{19}\,(\text{banjo}) = 190,$$

hence

$$4\sqrt{19}\,(\text{banjo}) = 190 - 2\sqrt{19},$$

$$\text{banjo} = \frac{190 - 2\sqrt{19}}{4\sqrt{19}} = \frac{95 - \sqrt{19}}{2\sqrt{19}}.$$

Now evaluate the final expression:

$$\text{accordion} + \text{banjo} \cdot \text{guitar} = 12 + \left(\frac{95 - \sqrt{19}}{2\sqrt{19}}\right)\sqrt{19}.$$

So

$$? = 12 + \frac{95 - \sqrt{19}}{2} = \frac{119 - \sqrt{19}}{2}.$$

Therefore,

$$? = \frac{119 - \sqrt{19}}{2} \approx 57.32.$$

2.3 Activity 1

1. Calculate the linear regression equation.
2. Create the scatter plot with the linear regression equation.

Test 1	10	18	13	6	8	5	12	15	15
Test 2	12	20	11	9	6	6	12	13	17

Solution

Take Test 1 as x and Test 2 as y. The data pairs are

$$(10, 12),\ (18, 20),\ (13, 11),\ (6, 9),\ (8, 6),\ (5, 6),\ (12, 12),\ (15, 13),\ (15, 17).$$

Since there are 9 pairs,

$$n = 9.$$

Now calculate the required sums directly from the table.

$$\sum x = 10 + 18 + 13 + 6 + 8 + 5 + 12 + 15 + 15 = 102$$

$$\sum y = 12 + 20 + 11 + 9 + 6 + 6 + 12 + 13 + 17 = 106$$

$$\sum x^2 = 10^2 + 18^2 + 13^2 + 6^2 + 8^2 + 5^2 + 12^2 + 15^2 + 15^2$$
$$= 100 + 324 + 169 + 36 + 64 + 25 + 144 + 225 + 225 = 1312$$

$$\sum xy = (10)(12)+(18)(20)+(13)(11)+(6)(9)+(8)(6)+(5)(6)+(12)(12)+(15)(13)+(15)(17)$$
$$= 120 + 360 + 143 + 54 + 48 + 30 + 144 + 195 + 255 = 1349$$

Substitute these values into the slope formula:

$$b = \frac{n\sum(xy) - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$b = \frac{9(1349) - (102)(106)}{9(1312) - 102^2}$$

$$b = \frac{12141 - 10812}{11808 - 10404}$$

$$b = \frac{1329}{1404}$$

$$b = \frac{443}{468} \approx 0.947$$

Now calculate the intercept:

$$a = \frac{\sum y - b\sum x}{n}$$

$$a = \frac{106 - \left(\frac{443}{468}\right)(102)}{9}$$

$$a = \frac{106 - \frac{15062}{468}}{9}$$

$$a = \frac{\frac{49608}{468} - \frac{15062}{468}}{9} = \frac{\frac{34546}{468}}{9}$$

$$a = \frac{34546}{4212} = \frac{17273}{2106} = \frac{737}{702} \approx 1.050$$

Therefore, the linear regression equation is

$$\hat{y} = a + bx$$

$$\hat{y} = 1.050 + 0.947x$$

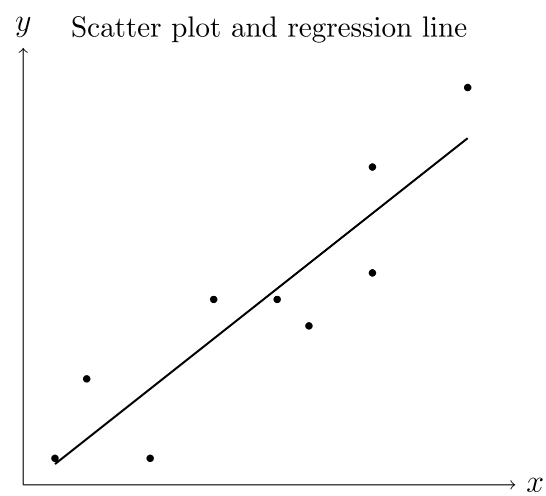
For the scatter plot, plot the nine points

$$(10, 12),\ (18, 20),\ (13, 11),\ (6, 9),\ (8, 6),\ (5, 6),\ (12, 12),\ (15, 13),\ (15, 17)$$

and then draw the regression line

$$\hat{y} = 1.050 + 0.947x.$$

The scatter plot with the regression line is shown below.



Here the plotted coordinates are the translated data points $(x - 4, y - 5)$, so the relative positions are preserved and the line shown corresponds to the same regression relationship.

2.4 Activity 1

1. Calculate the linear regression equation.
2. Create the scatter plot with the linear regression equation.

Test 1	10	18	13	6	8	5	12
Test 2	12	20	11	9	6	6	12

Solution

Take Test 1 as x and Test 2 as y. The data pairs are

$$(10, 12),\ (18, 20),\ (13, 11),\ (6, 9),\ (8, 6),\ (5, 6),\ (12, 12).$$

Since there are 7 pairs,

$$n = 7.$$

Now calculate the required sums directly from the table.

$$\sum x = 10 + 18 + 13 + 6 + 8 + 5 + 12 = 72$$

$$\sum y = 12 + 20 + 11 + 9 + 6 + 6 + 12 = 76$$

$$\sum x^2 = 10^2 + 18^2 + 13^2 + 6^2 + 8^2 + 5^2 + 12^2$$
$$= 100 + 324 + 169 + 36 + 64 + 25 + 144 = 862$$

$$\sum xy = (10)(12) + (18)(20) + (13)(11) + (6)(9) + (8)(6) + (5)(6) + (12)(12)$$
$$= 120 + 360 + 143 + 54 + 48 + 30 + 144 = 899$$

Substitute these values into the slope formula

$$b = \frac{n \sum (xy) - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$b = \frac{7(899) - (72)(76)}{7(862) - 72^2}$$

$$b = \frac{6293 - 5472}{6034 - 5184}$$

$$b = \frac{821}{850} \approx 0.966$$

Now calculate the intercept

$$a = \frac{\sum y - b \sum x}{n}$$

$$a = \frac{76 - \left(\frac{821}{850}\right)(72)}{7}$$

$$a = \frac{76 - \frac{59112}{850}}{7}$$

$$a = \frac{\frac{64600}{850} - \frac{59112}{850}}{7} = \frac{\frac{5488}{850}}{7}$$

$$a = \frac{5488}{5950} = \frac{2744}{2975} \approx 0.922$$

Therefore, the linear regression equation is

$$\hat{y} = a + bx$$

$$\hat{y} = 0.922 + 0.966x$$

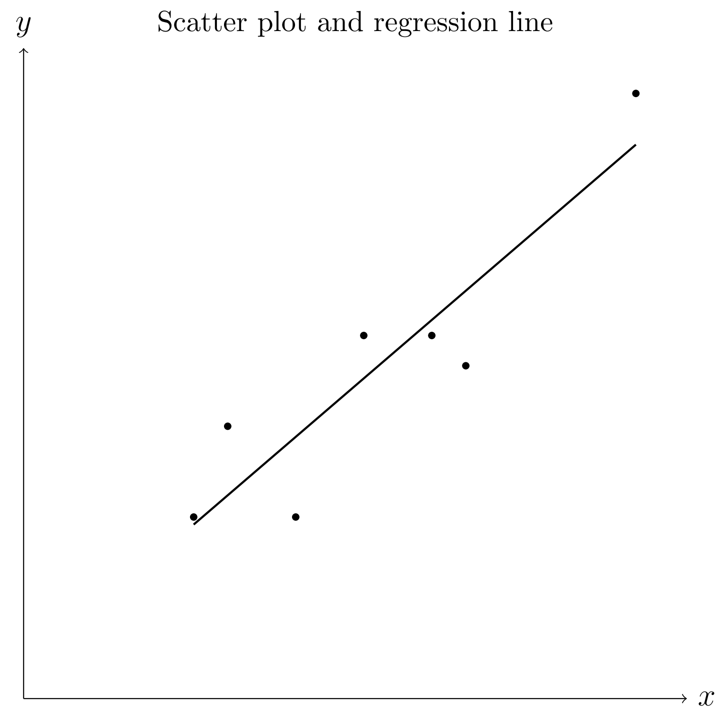
For the scatter plot, plot the seven points

$$(10, 12),\ (18, 20),\ (13, 11),\ (6, 9),\ (8, 6),\ (5, 6),\ (12, 12)$$

and then draw the regression line

$$\hat{y} = 0.922 + 0.966x.$$

The scatter plot with the regression line is shown below.



2.5 Activity 2 – Data set 2

1. Calculate the linear regression equation.

2. Create the scatter plot with the linear regression equation.

Minimum temperature (x)	17.7	19.8	23.3	22.4	22.0	22.0
Maximum temperature (y)	29.4	34.0	34.5	35.0	36.9	36.4

The table above shows the maximum and minimum temperatures (in °C) during a hot week in Melbourne.

Solution

For this data set,

$$n = 6, \qquad \sum x = 127.2, \qquad \sum y = 206.2,$$
$$\sum x^2 = 2717.98, \qquad \sum xy = 4394.03.$$

The slope is

$$b = \frac{6(4394.03) - 127.2(206.2)}{6(2717.98) - (127.2)^2} \approx 1.059.$$

The intercept is

$$a = \frac{206.2 - 1.058575445(127.2)}{6} \approx 11.925.$$

Therefore, the linear regression equation is

$$\hat{y} = 11.925 + 1.059x.$$

The scatter plot with the regression line is shown below.

Here the plotted coordinates are the translated data points $(x - 16, y - 28)$, so the relative positions are preserved and the line shown corresponds to the same regression relationship.

2.6 Activity 2 – Data set 3

1. Calculate the linear regression equation.
2. Create the scatter plot with the linear regression equation.

Balls faced	29	16	19	62	13	40	16	9	28	26	6
Runs scored	27	8	21	47	3	15	13	2	15	10	2

The table above shows the number of runs scored and the number of balls faced by batsmen in a 1-day international cricket match.

Solution

For this data set,

$$n = 11, \qquad \sum x = 264, \qquad \sum y = 163,$$

$$\sum x^2 = 8904, \qquad \sum xy = 5781.$$

The slope is

$$b = \frac{11(5781) - 264(163)}{11(8904) - 264^2} = \frac{1946}{2674} \approx 0.728.$$

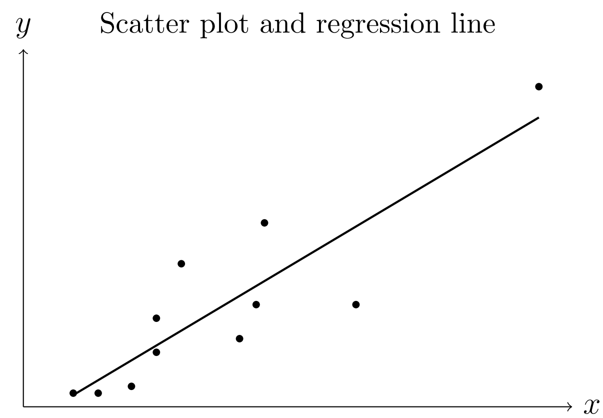
The intercept is

$$a = \frac{163 - 0.727803738(264)}{11} \approx -2.649.$$

Therefore, the linear regression equation is

$$\hat{y} = -2.649 + 0.728x.$$

The scatter plot with the regression line is shown below.



Here the plotted coordinates are the scaled data points $(x/5, y/5)$, so the relative positions are preserved and the line shown corresponds to the same regression relationship.